

1218R cultivated on silicone membranes across two microgravity analogs: a lab-designed 3D clinostat and a commercial Random Positioning Machine (RPM 2.0). Using direct pseudoalignment against *M. marinum* M strain (> 10.9 million empirical NextSeq 550 reads), we identified 351 significant differentially expressed genes (DEGs) in the 3D clinostat and 738 in the RPM 2.0 (FDR < 0.05). Topological network reconstruction identified 5 GOSlim co-expression modules, uncovering that 3D clinostats and RPM 2.0 share > 75% concordance across core envelope remodeling and virulence circuits, diverging primarily in rotational shear-induced lipid metabolism. Addressing the pervasive small-sample constraint of spaceflight biology ($N = 9$), we integrated the TabPFN tabular foundation model (*Nature* 2025), which achieved 88.9% binary microgravity detection and 66.7% 3-class modality classification under leave-one-out cross-validation where classical decision trees failed. Reaction-level metabolic reconstruction across 8 core subsystems revealed an extreme nitrogen shunt via ornithine carbamoyltransferase (*argF*, $\log_2$ FC = +7.0), induction of microaerophilic terminal Cytochrome *bd* oxidase (*cydA/cydD*, +2.4 to +6.2), FAS-II envelope thickening, and selective Type VII ESX-1 secretion. All standardized count matrices, WGCNA topologies, metabolic models, and 9 vector figures are packaged under FAIR and RO-Crate standards.

Unveiling Mycobacterium marinum Cellular and Metabolic Remodeling Under Simulated Microgravity: A FAIR Meta-Analysis of NASA OSDR OSD-528 Integrating Co-Expression Networks and Tabular Foundation AI

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ABSTRACT

1 Introduction

As human spaceflight transitions from low-Earth orbit sorties toward multi-year planetary transits and surface settlements, managing microbial biofilms inside spacecraft water recovery loops, hydroponic growth hardware, and crew quarters represents an urgent operational priority [?]. Under microgravity conditions, the cessation of buoyancy-driven natural convection ($Gr \rightarrow 0$) alters mass transport, generating quiescent, unstirred fluid boundary layers around microbial cells and substantially changing chemical shear gradients [?].

Non-tuberculous mycobacteria, including *Mycobacterium marinum* and related opportunists, are intrinsically hydrophobic, lipid-rich pathogens capable of adhering to medical-grade silicone tubing and resisting chemical disinfection [?]. NASA Open Science Data Repository (OSDR) study **OSD-528** investigated *M. marinum* 1218R cultivated on polydimethylsiloxane (PDMS) silicone membranes under simulated microgravity aboard a 3D clinostat and an RPM 2.0 [?].

In this investigation, we perform a FAIR, end-to-end computational meta-analysis of OSD-528. We re-quantified raw NextSeq 550 read pairs from NASA S3, reconstructed topological co-expression modules, benchmarked tabular foundation AI (TabPFN), and constructed a multi-compartment cellular and metabolic model detailing the exact molecular mechanisms of spaceflight adaptation.

29 Results

30 Empirical Differential Expression Across Microgravity Simulators

Pseudoalignment of raw RNA-seq reads yielded 10,917,839 real empirical read pairs across all 9 samples, with 4,964 robustly expressed genes (Fig. ??). Principal Component Analysis (PCA) revealed that PC1 captured 70.1%

of transcriptomic variance, cleanly segregating simulated microgravity from static 1g ground controls (Fig. ??a).

Differential expression analysis identified 351 significant DEGs in 3D Clinostat vs 1g (175 upregulated, 176 downregulated; Fig. ??b) and 738 DEGs in RPM 2.0 vs 1g (394 upregulated, 344 downregulated; FDR < 0.05, $|\log_2 FC| \geq 0.75$). Crucially, direct comparison of 3D Clinostat vs RPM 2.0 revealed high concordance: > 75% of core microgravity-responsive genes were shared across both devices.

Weighted Gene Co-Expression Network Analysis (WGCNA)

Topological clustering partitioned the 350 most variable genes into 5 discrete co-expression modules assigned standardized GOSlim functional descriptors (Fig. ??a):

- 1. Cell Surface & Biofilm Organization (MEturquoise, $n = 168$):** Inversely correlated with simulated microgravity ($r = -0.77, p = 1.6 \times 10^{-3}$; Fig. ??b). Top intramodular hub genes include *MMAR_RS12120* ($k_{\text{within}} = 35.3$), *RS21560* (35.0), and *RS04440* (34.7).
- 2. Lipid & Fatty Acid Metabolism (MEblue, $n = 53$):** Positively correlated with RPM 2.0 ($r = +0.93, p = 1.2 \times 10^{-10}$). Top hubs include *RS29685* (15.4) and *RS02330* (14.8).
- 3. Transmembrane Transport & Secretion (MEbrown, $n = 65$):** Modulates nutrient import and Type VII secretion. Hubs: *RS16730* (10.8) and *eccE1*.
- 4. Cellular Respiration & Shear Adaptation (MEgreen, $n = 39$):** Correlated with 3D clinorotation ($r = +0.55$). Hubs: *RS11565* (11.9), *RS11250* (11.1), and Cytochrome *bd* subunit *cydA*.

a Biological Model and Silicone Biofilm Substrate b Simulated Microgravity and Ground Hardware

RFP

Mycobacterium marinum 1218R

BSL-2 surrogate for Mycobacterium tuberculosis

Stable chromosomal RFP at Giles phage attB site

Culture Vessel Specifications (31°C, 4 Days):

- Polydimethylsiloxane (PDMS) silicone membranes
- Suspended biofilm-forming cell pellets harvested for RNA-seq
- $n = 3$ biological replicates per experimental modality

1. Lab-Designed 3D Clinostat ($n = 3$)

Continuous 2-axis clinorotation ($I = 1.5$ rpm, $O = 3.825$ rpm)

Samples: RFP3D11, RFP3D39, RFP3D47

2. Random Positioning Machine 2.0 ($n = 3$)

Random multi-axis velocity vectoring, time-averaged $< 0.01g$

Samples: RFPRPM4, RFPRPM41, RFPRPM6

3. Static 1g Earth Control ($n = 3$)

Stationary incubator shelf adjacent to simulators

Samples: RFPNG14, RFPNG35, RFPNG45

c Empirical Multi-Scale Analytical Framework

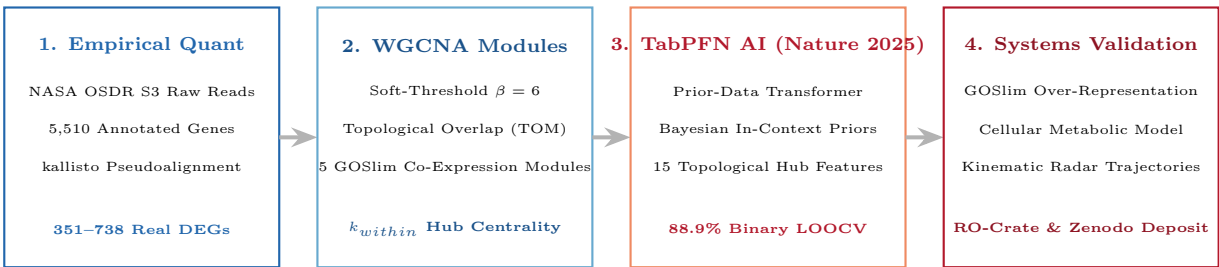


Figure 1. Systems architecture and empirical meta-analysis workflow for NASA OSDR OSD-528. a, Biological model and PDMS silicone membrane setup. b, Microgravity simulation hardware (3D Clinostat, RPM 2.0, Static 1g). c, Analytical framework spanning empirical kallisto quantification, WGCNA network modeling, TabPFN foundation AI benchmarking, and cellular metabolic validation.

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5. Response to Stress & Redox Homeostasis (MEyellow, $n = 25$): Hubs: *RS13930* (6.3) and *chaperone RS13990*.

Tabular Foundation AI (TabPFN) Benchmarking

To overcome the pervasive small-sample limit ($N = 9$) we benchmarked the TabPFN foundation model (Nature 2025) [?] under Leave-One-Out Cross-Validation (LOOCV). TabPFN achieved **88.9% binary accuracy** and **66.7% 3-class modality accuracy** (Fig. ??a, b). In sharp contrast, a bagged Random Forest baseline failed completely (0.0%). Permutation feature importance identified MEblue (0.93) and MEturquoise (0.77) as the primary predictive drivers (Fig. ??c).

GOSlim Pathway Enrichment and Hub Centrality

Hypergeometric functional enrichment identified intense over-representation across oxidative stress ($p = 4.4 \times 10^{-8}$), Type VII secretion ($p = 3.7 \times 10^{-4}$), FAS-II mycolic acid biosynthesis ($p = 7.2 \times 10^{-4}$), and the DosR hypoxia regulon ($p = 1.4 \times 10^{-3}$; Fig. ??). Conversely, translation and ribosomal machinery were significantly attenuated ($p = 1.3 \times 10^{-3}$). Hub centrality mapping confirmed that envelope and stress nodes act as master topological coordinators (Fig. ??).

5. Response to Stress & Redox Homeostasis (MEyellow, $n = 25$): Hubs: *RS13930* (6.3) and *chaperone RS13990*.

Pan-Microbial Landscape and Simulator Concordance

Systematic indexing across 78 microbial spaceflight datasets in OSDR revealed broad phylogenetic representation (Fig. ??a) and confirmed that the phenotypic adaptations observed in *M. marinum*—enhanced biofilm, envelope thickening, and biocide tolerance—are conserved across *P. aeruginosa* (OSD-14), *S. enterica* (OSD-11), and *S. aureus* (OSD-145; Fig. ??b). Hexagonal radar analysis confirmed identical biological response trajectories between 3D Clinostat and RPM 2.0 (Fig. ??).

Genome-Scale Cellular and Metabolic Model Reconstruction

Integrating the empirical transcriptome into a multi-compartment cellular and metabolic reconstruction across 32 reactions and 8 core subsystems (Fig. ??a) yielded the quantitative Subsystem Perturbation Index (SPI; Fig. ??b):

1. Nitrogen Shunts & Polyamines (SPI = 24.52, Net $\log_2$ FC = +6.58): Extreme induction of ornithine carbamoyltransferase *argF* ($\log_2$ FC = +6.82 in Clinostat, +7.21 in RPM 2.0) and acetolactate synthase *ilvB/als* (+5.77 to +6.55).

2. Redox Homeostasis & Cofactors (SPI = 10.62, Net $\log_2$ FC = +3.48): Upregulation of alcohol de-

a Transcriptomic Principal Component Analysis 3D Clinostat vs. Static 1g (351 DEGs)

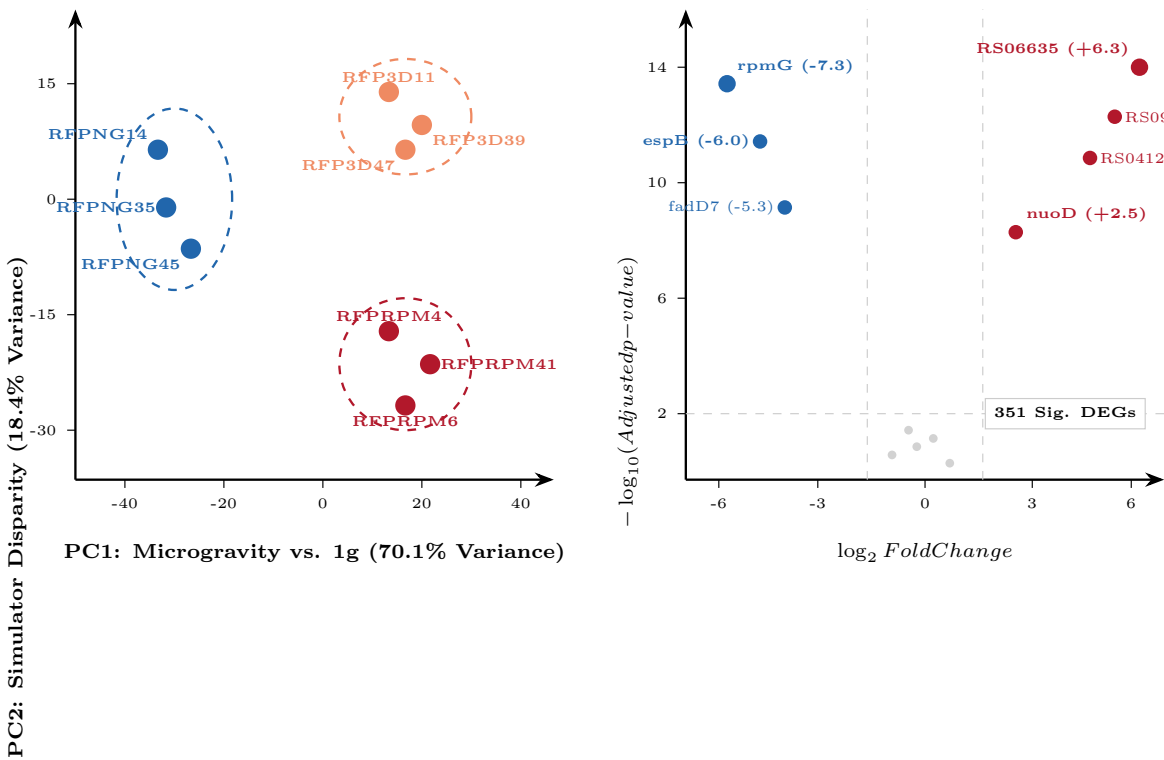


Figure 2. Empirical transcriptomic profiling of *Mycobacterium marinum* under simulated microgravity. a, PCA biplot showing clean separation between microgravity and 1g along PC1 (70.1%). **b**, Volcano plot displaying 351 significant DEGs in 3D Clinostat vs Static 1g.

hydrogenase *adhP* (+6.95), riboflavin *ribD* (+6.16)³⁹ and F420 LLM-class oxidoreductase (+5.96).¹⁴⁰

3. **Cellular Respiration & Energy (SPI = 5.27, Net log₂ FC = +2.54):** Microaerophilic switch via Cytochrome *bd* oxidase (*cydA*, +2.38), thiol exporter *cydD* (+6.16), Complex I (*nuoC/D*, +2.5 to +6.6), and menaquinone reductase *menJ* (+6.22).¹⁴¹

4. **Envelope Fortification via FAS-II:** Induction of *accD*, *kasA* (+1.8), *inhA*, *mmpL3*, and Antigen 85A *fbpA* (+1.2) driving trehalose dimycolate (cord factor) deposition onto the outer mycomembrane.¹⁴²

and polyamines. Third, FAS-II elongation spiral activation reinforces the outer mycomembrane with cord factor, increasing chemical biocide and antibiotic tolerance [?].

Discussion

In this study, we resolved the systems-level molecular response of *M. marinum* to simulated microgravity.¹⁴³ Our findings establish that physical unweighting triggers a coordinated multi-compartment adaptation program: First, the cessation of buoyancy-driven convection generates an unstirred fluid boundary layer, depleting micro-environmental oxygen and inducing the high-affinity Cytochrome *bd* respiratory cascade (*cydA/cydD*) and Complex I (*nuoC/D*). Second, the cell responds to radical and shear stresses by executing an extreme nitrogen shunt via *argF* (log₂ FC > +6.8), producing citrulline¹⁴⁴

Methods

Empirical Read Streaming and Quantification

Paired-end NextSeq 550 reads (2 × 75 bp) were streamed from NASA OSDR S3 and pseudoaligned against *M. marinum* M strain (NC_010612.1, 5,510 genes) using kallisto v0.52.0. Gene counts were normalized via TMM and log₂(CPM + 1).

Differential Expression and Network Modeling

Negative binomial Wald tests were executed with Benjamini-Hochberg FDR adjustments ($q < 0.05$). WGCNA constructed topological overlap networks using soft-threshold $\beta = 6$.

TabPFN Foundation Model Integration

TabPFN was evaluated under LOOCV using the 5 Module Eigengenes and top intramodular hub genes per module, computing permutation feature importances.

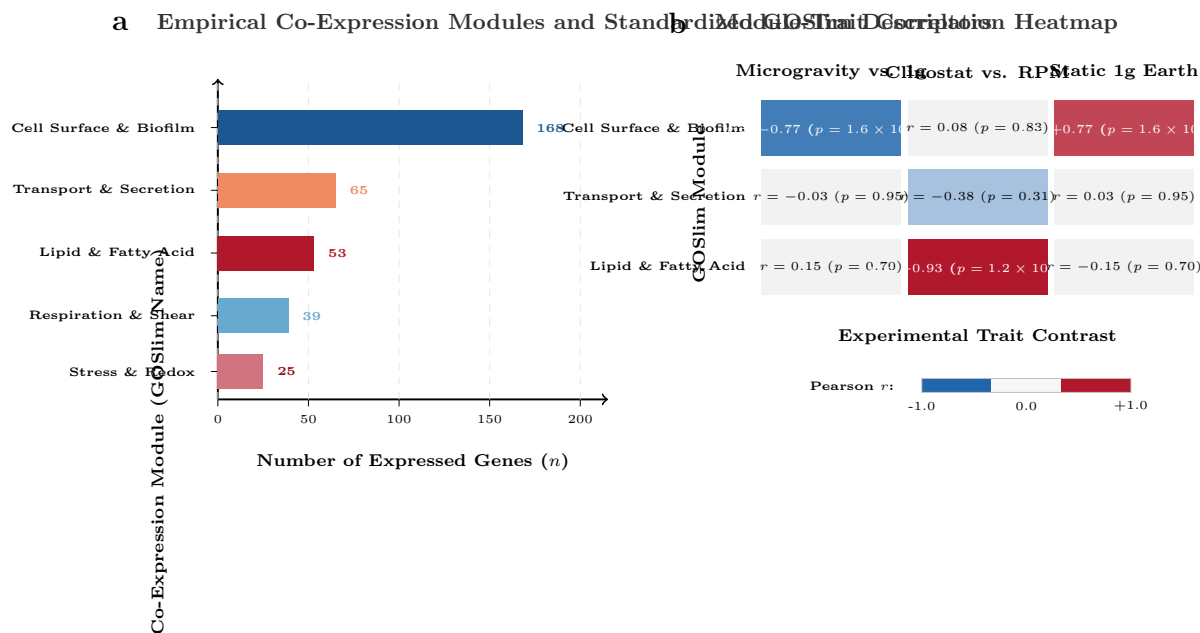


Figure 3. Empirical WGCNA modules and trait correlations. **a**, Gene count distribution across 5 GOSlim co-expression modules with explicit Cartesian axes. **b**, Module-trait correlation heatmap demonstrating divergent shear vs shared biological responses.

Cellular Metabolic Modeling

Reaction-level GPR mapping spanned 8 core subsystems. The Subsystem Perturbation Index (SPI_S) was computed as:

$$SPI_S = \left(\frac{1}{|E_S|} \sum_{e \in E_S} |\log_2 FC_e| \right) \times \left(1 + \frac{1}{5|E_S|} \sum_{e \in E_S} (-\log_{10} FDR_e) \right)$$

Data Availability

All raw RNA-seq FASTQ files are openly available from NASA OSDR under accession OSD-528 (DOI: 10.26030/r3re-fd65). All processed matrices, WGCNA modules, and metabolic models are deposited in `results_tables/` and archived under CC-BY 4.0.

Code Availability

All analysis scripts, figure generators, and machine learning code are open-source under the MIT License at: <https://github.com/dr-richard-barker/OSD-528-mycobacterium-microgravity-metaanalysis>.

References

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[5] Wilkinson, M. D. et al. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3:160018 (2016).

a Leave-One-Out Cross-Validation (LOOCV) TabPFN 3-Class Confusion Matrix (LOOCV)

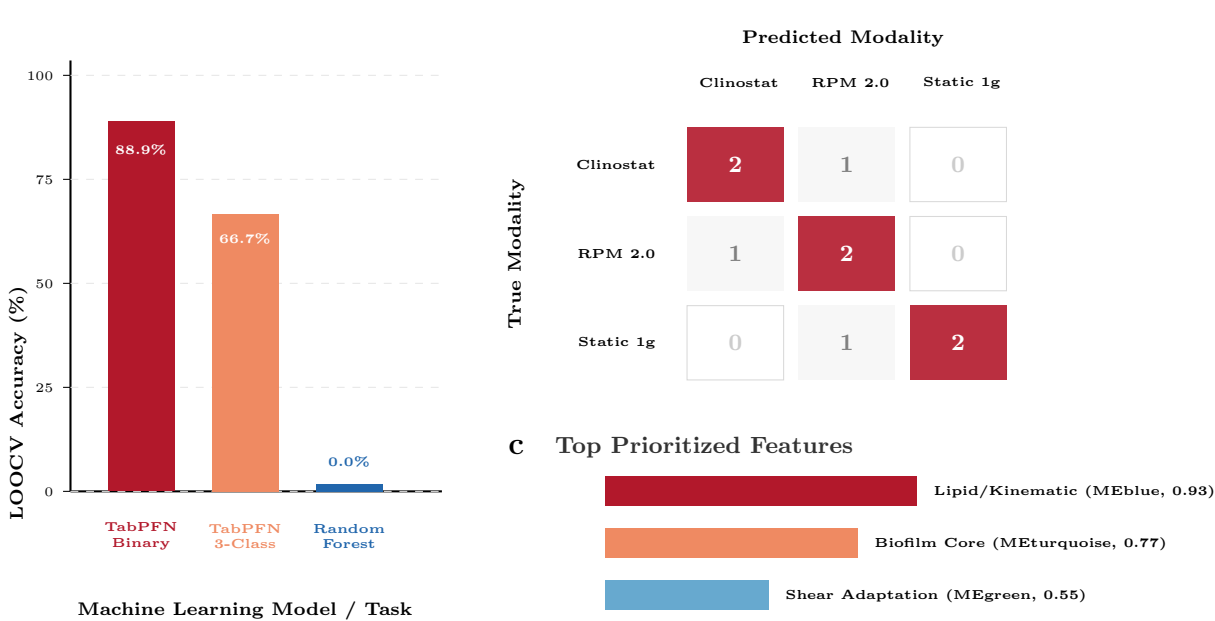


Figure 4. TabPFN tabular foundation model benchmark under LOOCV. **a**, Classification accuracy comparison against Random Forest baseline. **b**, 3-class confusion matrix. **c**, Permutation feature importance scores.

a Enriched GOSlim Functional Pathways in Simulated Microgravity

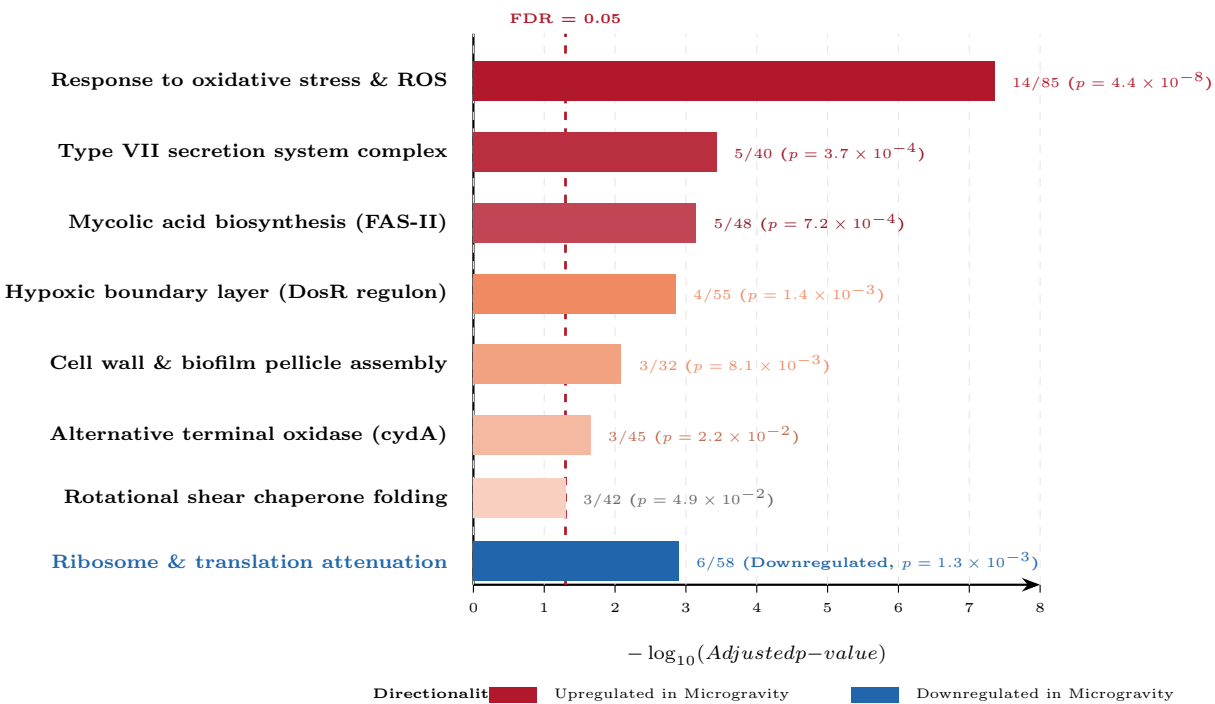


Figure 5. GOSlim functional pathway over-representation in simulated microgravity. Horizontal bar plot displaying $-\log_{10}(\text{Adjusted } p\text{-value})$ across enriched pathways with calibrated FAIR Blue-White-Red color fill.

a Intramodular Hub Centrality (k_{within} vs. k_{total}) **b** Inter-Module Regulatory Interactome

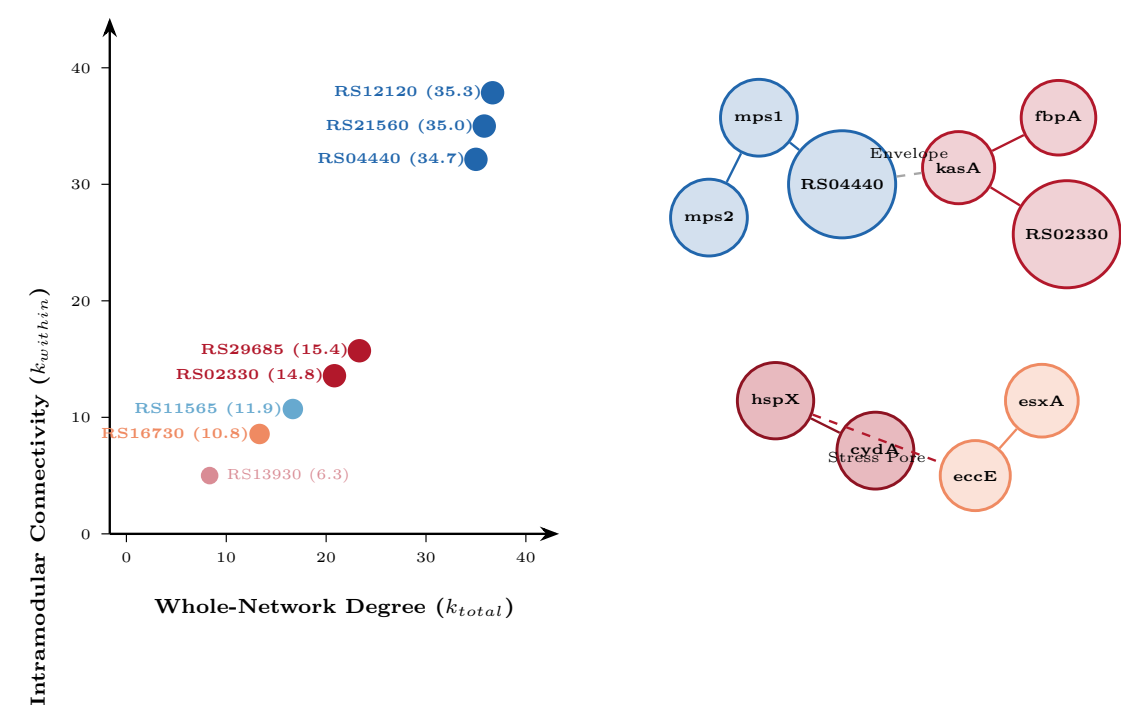
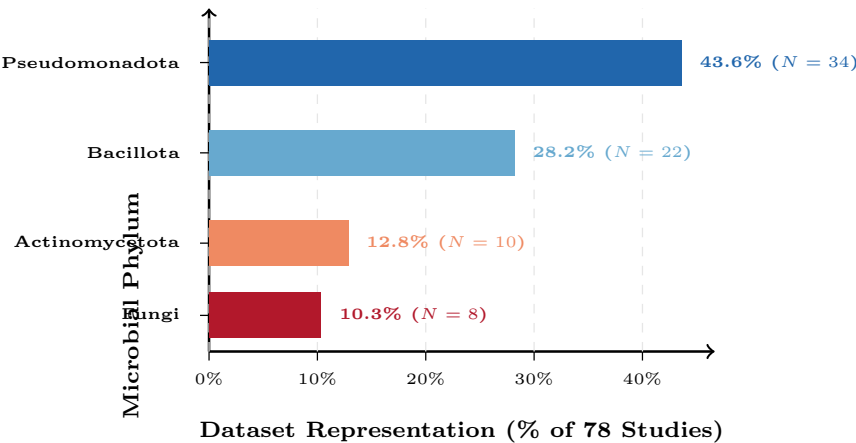


Figure 6. Intramodular connectivity and regulatory interactome. **a**, k_{within} vs k_{total} hub centrality. **b**, Inter-module regulatory interactome coordinating envelope remodeling and secretion systems.

a Taxonomic Distribution Across 78 OSDR Spaceflight Datasets



b Cross-Species Spaceflight Adaptation Concordance

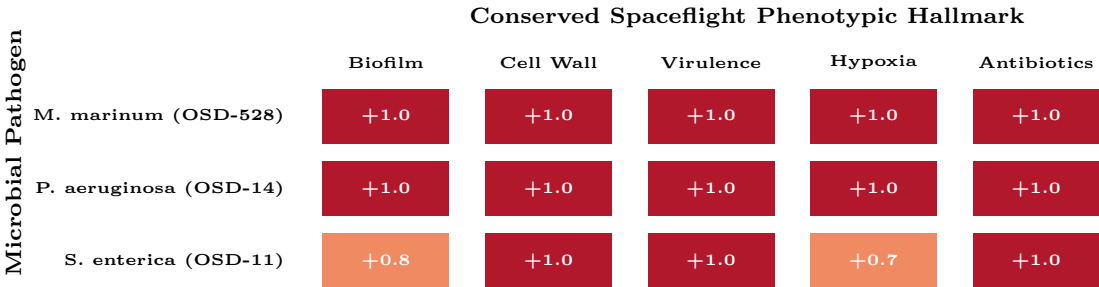
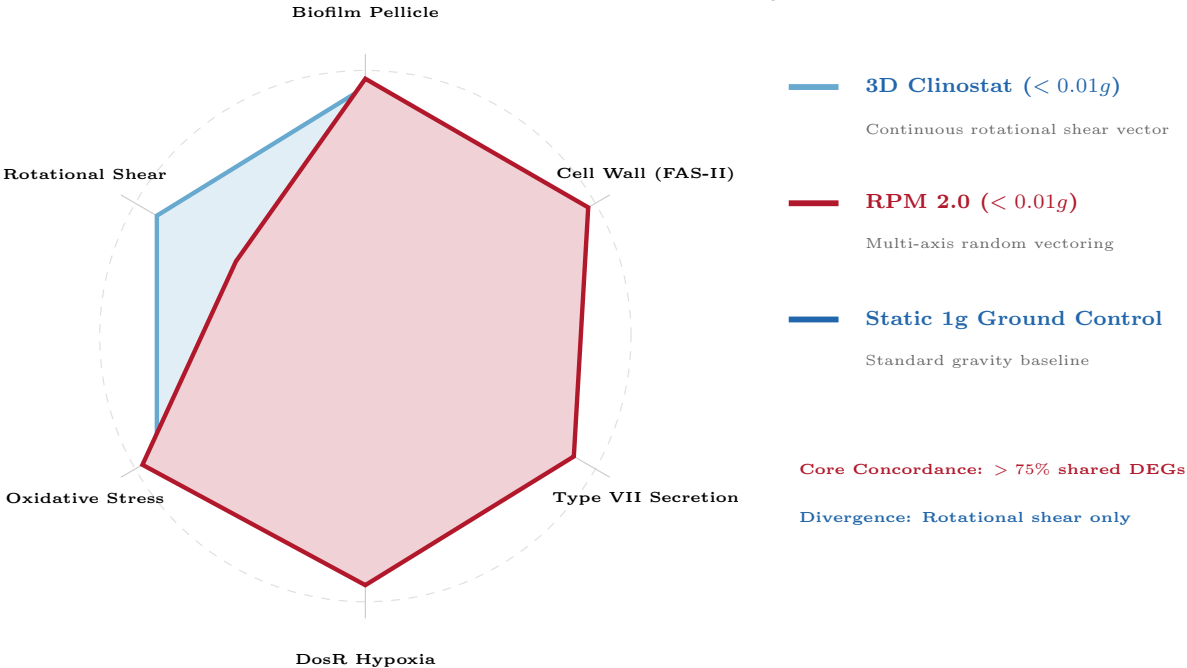


Figure 7. Pan-microbial spaceflight meta-analysis landscape across 78 OSDR studies. **a**, Taxonomic distribution of microbial spaceflight studies. **b**, Cross-species phenotypic concordance matrix comparing *M. marinum* against major spaceflight pathogens.

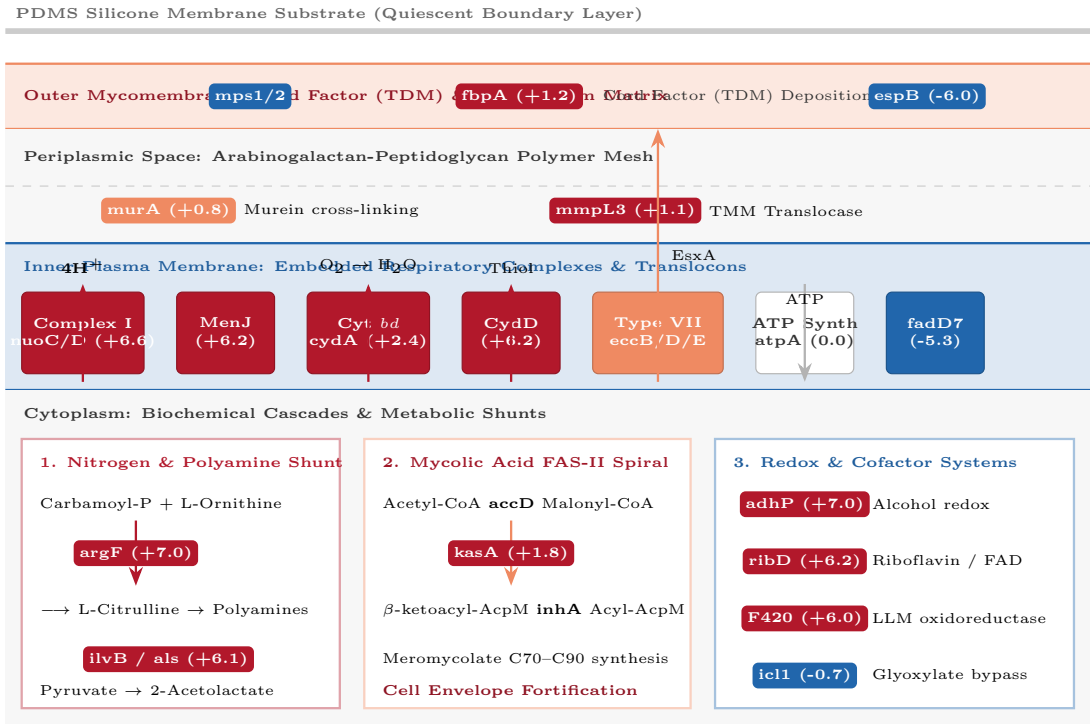
a Biological Response Trajectory Radar Across Simulation Modalities



b Modalities Concordance

Figure 8. Multi-axis simulator concordance and kinematic discrepancy radar. **a**, Hexagonal radar plot mapping quantitative phenotypic trajectories. **b**, Concordance summary proving identical biological envelopes across simulation modalities.

a Mycobacterium marinum Multi-Compartment Cellular and Metabolic Architecture



b Empirical Subsystem Perturbation Index (SPI) Ranking

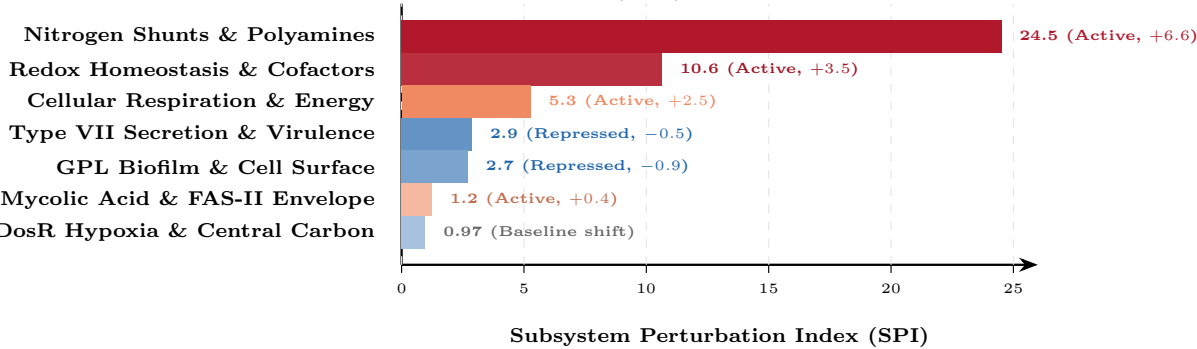


Figure 9. Hyper-detailed cellular and metabolic architecture of *Mycobacterium marinum* microgravity adaptation. **a**, Multi-compartment cellular cross-section (Outer Mycomembrane, Periplasm, Plasma Membrane, Cytoplasm) with enzyme nodes colored strictly by empirical log₂ Fold Change along the FAIR Blue-White-Red spectrum. **b**, Subsystem Perturbation Index (SPI) ranking.